RL with KL penalties is better viewed as Bayesian inference

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Abstract

Reinforcement learning (RL) is frequently employed in fine-tuning large language models (LMs) to penalize them for undesirable features of generated sequences, such as offensiveness or harmfulness. In this paper, we analyze challenges associated with treating language models as RL policies and show how avoiding those challenges requires moving beyond the RL paradigm.

We start by observing that naïve maximisation of the standard RL objective unavoidably leads to distribution collapse: turning the LM into a degenerate distribution. Then, we analyze KL-regularised RL, a widely used recipe for fine-tuning LMs, which additionally constrains the fine-tuned LM to stay close to its original distribution. We show that KL-regularised RL is equivalent to variational inference: approximating a Bayesian posterior which specifies how to update a prior LM to conform with evidence provided by the reward function.

Based on these observations, we sketch a Bayesian perspective that provides a first-principles derivation for KL-regularised RL. The Bayesian perspective also separates the modelling problem (defining a target distribution specifying the desired behaviour of an LM) and the inference problem (approximating that target distribution). Finally, it suggests that RL is not a good formal framework for thinking about fine-tuning LMs.

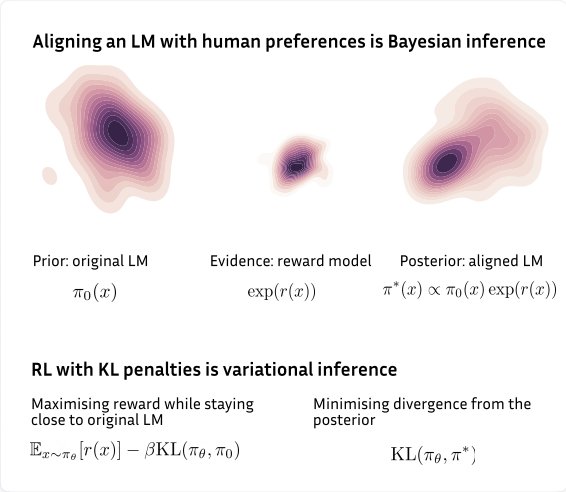

Figure 1: In the paper, we argue that aligning language models (LMs) with human preferences is a Bayesian inference problem and RL with KL penalties corresponds to solving it via variational inference.

1 Introduction

Large language models (LMs), such as GPT-3 (Brown et al., 2020), tend to generate outputs that reflect undesirable features of their training data such as offensiveness (Gehman et al., 2020), social bias (Bender et al., 2021), harmfulness (Bai et al., 2022) or dishonesty (Lin et al., 2021). Addressing these biases and constraining LMs to be honest, helpful and harmless is an essential part of the problem of aligning LMs with human preferences (Askell et al., 2021). One intuitive approach to aligning LMs is reinforcement learning (RL): capturing human preferences as a reward function and fine-tuning the LM to maximise the reward expected under LM distribution. A practical recipe for implementing this idea is RL from human feedback (Ziegler et al., 2019): first, a reward model is trained to predict which of two texts a human prefers and then a pretrained LM is fine-tuned to maximise reward given by the reward model while also being penalized for Kullback-Leibler (KL) divergence from its initial distribution. However, despite immense popularity of RL from human feedback (Stiennon et al., 2020; Ouyang et al., 2022; Perez et al., 2022; Bai et al., 2022), the motivation for KL penalty is not widely understood.

In this paper, we discuss an underappreciated perspective on KL-regularised RL — the objective employed by RL from human feedback for fine-tuning LMs — which explains its empirical

success. We start with describing a problem that arises from naively applying the standard RL objective: distribution collapse. The optimal policy under the RL objective would be a minimal-entropy LM generating a small set of sequences that obtain the highest reward. Then, we discuss how KL-regularised RL avoids distribution collapse due to its KL penalty. This constraint, we argue, transforms the problem from RL to Bayesian inference: updating a prior to conform with evidence provided by the reward. The Bayesian perspective moves KL-regularised RL closer to other divergence-minimisation-based approaches to fine-tuning LMs (Khalifa et al., 2021) and, more broadly, to other divergence-minimisation-based accounts of control (Levine, 2018; Hafner et al., 2020). These divergence minimisation approaches naturally avoid the distribution collapse problem because they formalize the agent as a generative model. In contrast, RL avoids distribution collapse only with reward functions that make it equivalent to divergence minimisation. Therefore, we conclude, RL is not an adequate formal framework for problems such as fine-tuning LMs.

2 Fine-tuning language models using standard RL and distribution collapse

Let $\mathcal{X}$ be the set of sequences of tokens from some vocabulary. An LM π can be seen as a probability distribution over $\mathcal{X}$. While most modern LMs are autoregressive, for simplicity we will only talk about full sequences, e.g. $\pi(x)$ denotes the probability of a sequence $x \in \mathcal{X}$. Similarly, a reward function r assigns sequences $x \in \mathcal{X}$ with scalar rewards. In practice, $r(x)$ could represent human preferences we would like π to be aligned with, e.g. a non-offensiveness reward would assign low values to sequences that are offensive.

If π_θ is our parametric LM (with parameters θ), the RL objective for fine-tuning it with our reward function r is just the reward expected under LM distribution:

$$J_{\text{RL}}(\theta) = \mathbb{E}_{x \sim \pi_\theta} r(x). \tag{1}$$

Intuitively, maximising $J_{\text{RL}}(\theta)$ means sampling a number of sequences from the LM and rewarding the LM for good sequences and penalising for bad ones (e.g. offensive sentences).

The problem with the RL objective is that it treats the LM as a policy, not as a generative model. While a generative model is supposed to capture a diverse distribution of samples, a policy is supposed to chose the optimal action. Since we don't have a notion of state for LMs, the RL objective reduces to searching for x^*, the sequence with highest reward. If there is one, the optimal policy π^* is a degenerate, deterministic generative model that puts entire probability mass on that single sequence:

$$\pi^* = \text{argmax}_\theta J_{\text{RL}}(\theta) = \delta_{x^*}, \tag{2}$$

where δ_{x^*} is a Dirac delta distribution centred on x^*. If there are multiple optimal sequences x^*, probability mass would be put only on them.

This failure mode is not purely theoretical. Empirically, distribution collapse induced by maximising reward manifests as decreased fluency and diversity of samples from the LM, which can be measured in terms of perplexity, entropy and the frequency of repetitions. Degeneration of this kind was observed in multiple language generation tasks ranging from translation (Choshen et al., 2019), summarisation (Paulus et al., 2018), story generation (Tambwekar et al., 2019), video captioning (Pasunuru and Bansal, 2017), dialogue (Jaques et al., 2019), to code generation (Korbak et al., 2021) and LM debiasing (Khalifa et al., 2021).

While the distribution collapse problem is exacerbated by RL failure modes such as insufficient exploration or reward hacking, it is distinct from exploration-exploitation trade-off or reward misspecification. Even with perfect exploration (if we sampled sequences uniformly from $\mathcal{X}$ as opposed to sampling from π_θ), the optimal policy will still put all probability mass on x^*. Similarly, even if r perfectly captures human preferences across the whole space of possible sequences $\mathcal{X}$ and if x^* is truly the best thing, we still would not want the LM to generate only x^*.[1] Essentially, the distribution collapse problem arises from the fact that

[1] There is a case to be made that in conditional generation (e.g. translation or summarisation) one really cares *only* about the single best output for a given context (e.g. summary of a document). There are still, however, substantial benefits of caring about distributional aspects in conditional generation. First, when the LM produces a full distribution, we are able to measure its uncertainty. For larger models, these uncertainty estimates happen to be well-calibrated and allow for safer deployment in high-stakes scenarios (Kadavath et al., 2022). Second, MAP estimates of the output distribution (the single, most likely output) are frequently of poor quality and can be substantially improved upon with decoding procedures taking into account the entire distribution, e.g. minimum Bayes risk in translation (Eikema and Aziz, 2020) or self-consistency chain-of-thought in question answering (Wang et al., 2022). Dohan et al. (2022) provide a unifying perspective on multi-step generation as latent variable modelling.

the RL objective for LM alignment is flawed: it does not care about preserving distributional properties of an LM and will always penalise the LM for putting any probability mass on non-optimal sequences until the LM collapses into a degenerate distribution.

3 Fine-tuning language models via KL-regularised RL

There is an obvious solution to the distribution collapse problem: including preserving distributional properties of an LM as part of the reward function. The notion of preserving distributional properties of an LM π_θ can be formalised as penalising for Kullback-Leibler (KL) divergence between π_θ and some other, pretrained LM π_0. Typically, π_θ is initialised to π_0 and then fine-tuned to maximise the following objective:

$$J_{\text{KL-RL}}(\theta) = \mathbb{E}_{x\sim\pi_\theta}[r(x)] - \beta D_{\text{KL}}(\pi_\theta, \pi_0). \quad (3)$$

The first term in the right-hand side of (3) is equivalent to $J_{\text{RL}}(\theta)$ in (1) while the second additionally constrains π_θ to stay close (in terms of KL) to π_0. Almost always some reward needs to be sacrificed for that; the coefficient β determines the trade-off of how much reward is needed to justify departing from π_0 by a certain distance. This objective is commonly used as part of a popular recipe for fine-tuning LMs termed "RL from Human Feedback" (RLHF) and works surprisingly well in practice (Ziegler et al., 2019; Stiennon et al., 2020; Perez et al., 2022; Bai et al., 2022). Earlier approaches to fine-tuning LMs employing this objective used the called it "conservative fine-tuning" (Jaques et al., 2017) or KL-control (Jaques et al., 2019). Here, we focus only on the policy optimisation part of this setup, which we term "KL-regularised RL".

The KL-regularised RL objective (3) can easily be reformulated as just expected reward as in (1). We only have to define a new reward function $r'_\theta(x)$ which incorporates the original reward r and the KL penalty, using the definition of KL divergence:

$$J_{\text{KL-RL}}(\theta) = \mathbb{E}_{x\sim\pi_\theta}[r'_\theta(x)], \text{where}$$
$$r'_\theta(x) = r(x) + \beta(\log \pi_0(x) - \log \pi_\theta(x)). \quad (4)$$

But is framing the maximisation of (4) as RL really necessary? In the next section, we will develop an alternative view of this objective – as an approximate solution to a Bayesian inference problem – and argue that it is a more appealing framing.

4 KL-regularised RL as variational inference

Fine-tuning a pretrained LM π_0 to align with preferences encoded by a reward function r is essentially a Bayesian inference problem. Intuitively, Bayesian inference is the problem of updating a distribution to conform with new evidence. In our setting, we're updating π_θ, which is initially equal to a prior π_0 to conform with evidence provided by the assumption that π_θ is optimal in terms of r. A reward function can be represented as a distribution over $\mathcal{X}$ that makes high-reward sequences more likely that low-reward sequences. A simple way of doing that is exponentiating reward r and renormalizing it. Then, the posterior is given by:

$$\pi^*_{\text{KL-RL}}(x) = \frac{1}{Z}\pi_0(x)\exp(r(x)/\beta), \quad (5)$$

where π_0 is the prior, $\exp(r(x)/\beta)$ is the evidence provided by the reward function (scaled by temperature β) and Z is a constant ensuring that $\pi^*_{\text{KL-RL}}$ is a normalised probability distribution. $\pi^*_{\text{KL-RL}}$ represents a version of π_0 updated to account for the reward r. As we demonstrate in the Appendix, it also happens to coincide with the optimal policy for $J_{\text{KL-RL}}$:

$$\pi^*_{\text{KL-RL}} = \text{argmax}_\theta J_{\text{KL-RL}}(\theta) \quad (6)$$

Moreover, the KL-regularised RL objective can be cast as minimising the KL divergence between the LM π_θ and this target distribution $\pi^*_{\text{KL-RL}}$:

$$J_{\text{KL-RL}}(\theta) \propto -D_{\text{KL}}(\pi_\theta, \pi^*_{\text{KL-RL}}) \quad (7)$$

This divergence is different from the KL penalty term $D_{\text{KL}}(\pi_\theta, \pi_0)$ in (3). Minimising this new divergence coincides with a variational inference (Blei et al., 2017), a well-known approach to approximating Bayesian inference. More formally, $J_{\text{KL-RL}}(\theta)$ is the evidence lower bound (ELBO) on the log likelihood of π_θ being optimal under r, assuming a prior π_0. Minimising this bound makes π_θ approximate the true posterior $\pi^*_{\text{KL-RL}}$. A derivation of these equalities can be found in the Appendix below.

Why is this picture insightful? It explains where the KL penalty term $\beta D_{\text{KL}}(\pi_\theta, \pi_0)$ in KL-regularised RL's original objective comes from. It is necessary to transform the problem from RL to minimising a divergence from a target distribution π^*_{RLKL}. This in turn makes the distributional character of an LM a first-class citizen which explains

why KL-regularised RL is able to maintain the fluency and diversity of the original LM π_0.

5 Separation of modelling and inference

The Bayesian perspective suggests that that aligning an LM with task preferences is a two-step process. It consists of, first, defining a distribution specifying the desired behaviour of your LM, and second, solving the problem of sampling from that posterior. These two steps roughly correspond to modelling and inference in probabilistic programming (Goodman and Stuhlmüller, 2014). Modelling is encoding knowledge in probabilistic terms (usually by defining a probabilistic graphical model) while inference corresponds to using this model to answer queries. It is hard to overstate how useful — theoretically and practically — separating these two concerns could be. Let us discuss these two steps, separately, below.

Modelling The LM is natively a probability distribution and autoregressive models allow for both sampling and evaluating likelihoods. Therefore, most modelling decisions are usually around interpreting task preferences in probabilistic terms. Turning a reward function r into a distribution by exponentiating it ($\frac{1}{Z}\exp(r(x))$ is the standard approach, but there are others. In some cases, task preferences can be binary, for instance a dialogue system might be required to *never* generate a curse word (but is free to behave normally otherwise). Then, following (Khalifa et al., 2021), one could define $\pi^*(x) = \frac{1}{Z}\pi_0(x)b(x)$, where $b(x) = 1$ if x contains a curse and 0 otherwise. Then, sequences x containing curses have probability zero according to π^* (hence π^* is non-cursing) but all other strings keep the original probability $\pi_0(x)$ up to Z (hence no degeneration).

Inference The posteriors mentioned above are generally non-parametric: they might lie outside the class of probability distributions representable by parametric LMs. Designing an algorithm able to generate samples matching this posterior distribution constitute the inference problem. Broadly, there are two families of algorithms for inference on probabilistic graphical models: variational inference and sampling-based approaches. Variational inference tries to find the set of weights θ that give rise to a distribution π_θ closest (in terms of KL) to the true posterior. Sampling-based techniques, such as MCMC (Brooks et al., 2011), do not represent the true posterior explicitly, but compute samples from a distribution resembling the true posterior. In the previous section, we have shown that KL-regularised RL corresponds to inference via variational inference. But sampling-based inference algorithms also have analogues for LMs in decoding-time methods. Decoding-time methods boil down to simulating a posterior, aligned LM π^* by modifying the generation procedure applied on top of the original LM π_0. The simplest example of that is filtering (also known as rejection sampling): if the LM generates an unacceptable sample, it is discarded and a new sample is generated (Xu et al., 2020). More elaborate decoding-time methods include weighted decoding (See et al., 2019) or PPLM (Dathathri et al., 2019).

To summarise, the Bayesian view provides a unifying perspective on fine-tuning and decoding-time approaches to LM alignment. They mirror variational inference and sampling-based inference algorithms for probabilistic graphical models. But a more fundamental advantage, to our mind, is the separation of concerns between defining a desired behaviour of an LM and approximating it. The choice of posterior is independent of how it is going to be approximated. This, in turn, separates two failure modes: misspecifying the model (i.e. not capturing task preferences) and failing to approximate the model well enough.

6 Is RL a good framework for fine-tuning language models?

There is a family of other divergence minimisation approaches to fine-tuning LMs which are not equivalent to RL. Take Generative Distributional Control (GDC) (Khalifa et al., 2021; Korbak et al., 2022a), an approach to fine-tuning LMs that obtains results comparable with KL-regularised RL but minimises a slightly different divergence (forward as opposed to reverse KL). However, $J_{\text{GDC}}(\theta)$ is no longer equivalent to RL (Korbak et al., 2022b) because the expectation in forward KL divergence is with respect to a $\pi^*_{\text{KL-RL}}$, not π_θ. Similarly, standard supervised training objective can be seen as minimising $D_{\text{KL}}(\pi^*_{\text{MLE}}, \pi_\theta)$, a divergence from the empirical distribution π^*_{MLE} provided by the training set.

One can therefore mount a double dissociation argument in favour of the divergence minimisation perspective on KL-regularised RL: RL without KL divergence minimisation leads to degenera-